NATIONAL RESEARCH CONFERENCE ON

CONVERGING AI AND BIG DATA FOR CYBERSECURITY EXCELLENCE

(NCCABCE-2024)

ORGANIZED BY

DEPARTMENT OF IT, SD, MSC.BDA

JAI HIND COLLEGE, CHURCHGATE MUMBAI

COMPARATIVE ANALYSIS OF TWOFISH AND BLOWFISH FOR SECURITY IN HYBRID CLOUD MODELS

Paper ID: (NRC6A20)

INTRODUCTION

- Cloud computing has become essential for modern organizations, offering scalable and flexible data management solutions.
- Hybrid Cloud integrates private and public cloud systems, balancing security, scalability, and cost-efficiency.
- However, the movement of sensitive data between private and public environments introduces significant security challenges.
- Data encryption ensures confidentiality and integrity, making it a critical component of hybrid cloud security.
- This study compares Twofish and Blowfish encryption algorithms to evaluate their suitability for securing data in hybrid cloud environments.
- The analysis focuses on security strength, encryption performance, and adaptability to hybrid cloud requirements.

PROBLEM STATEMENT

- Hybrid cloud models enable seamless data flow between private and public environments, enhancing flexibility and scalability.
- However, this integration exposes sensitive data to increased risks of unauthorized access, data breaches, and tampering during transfer and storage.
- The challenge lies in implementing encryption algorithms that:
  - Provide **strong security** to protect sensitive data.
  - Maintain **high performance** without compromising system efficiency.
  - **Adapt** to the dynamic and scalable nature of hybrid cloud architectures.
- There is a need to identify and compare encryption algorithms that balance security robustness and operational efficiency for hybrid cloud systems.

LITERATURE REVIEW

- Hybrid cloud environments demand encryption algorithms that ensure data confidentiality, integrity, and scalability across public and private cloud segments.
- Twofish Algorithm:
  - High security with **128-bit** block size and key sizes up to **256 bits**.
  - Well-suited for high-security environments but incurs computational overhead.
  - Resistant to advanced cryptanalytic attacks, suitable for large datasets.
- Blowfish Algorithm:
  - Fast encryption and decryption with a smaller **64-bit** block size.
  - Efficient for small datasets and resource-constrained environments.
  - Vulnerable to birthday attacks on larger datasets, limiting its scalability.
- Existing studies highlight the **trade-off between security and performance**, emphasizing the importance of selecting the right encryption algorithm based on specific hybrid cloud requirements.

METHODOLOGY/PROCESS-FLOW

- Literature Collection:
  - Reviewed academic papers, journals, and industry reports on hybrid cloud encryption challenges.
  - Focused on studies comparing Twofish and Blowfish algorithms.
- Key Evaluation Parameters:
  - Security robustness: Resistance to attacks (e.g., brute force, differential cryptanalysis).
  - Encryption and decryption speed: Efficiency in handling small vs. large datasets.
  - Scalability: Adaptability to hybrid cloud's dynamic data requirements.
  - Resource consumption: Memory and computational overhead.

METHODOLOGY/PROCESS-FLOW

- Synthesis and Analysis:
  - Compared strengths, limitations, and trade-offs in security vs. performance.
  - Identified best-fit scenarios for each algorithm in hybrid cloud environments.
- Outcome:
  - Established insights into Twofish and Blowfish's roles in securing hybrid cloud systems.

RESULTS AND DISCUSSION

Comparison of Twofish and Blowfish:

- **Security**:
  - **Twofish** offers superior **security strength** due to its larger block size and complex key schedule, making it more resistant to cryptanalytic attacks.
  - **Blowfish**, while still secure, is more vulnerable to attacks on larger datasets due to its smaller 64-bit block size.
- **Performance**:
  - **Blowfish** excels in terms of **encryption speed** and is more efficient for **smaller datasets** and **low-resource environments**.
  - **Twofish** is better suited for **larger datasets** but incurs higher computational overhead, affecting **speed** in less resourceful systems.
- **Scalability**:
  - **Twofish** is more **scalable** and performs better in **large-scale hybrid cloud environments** that require dynamic data handling across private and public clouds.
  - **Blowfish**, with its smaller block size, is less efficient for large datasets, making it **less scalable** in large hybrid cloud systems.

RESULTS AND DISCUSSION

Best Fit Applications:

- Twofish:
  - Best for **high-security applications** with **large datasets**, such as **financial systems**, **enterprise data storage**, and **big data analytics** in hybrid clouds.
  - Ideal for organizations with strict **compliance** requirements and need for strong encryption across both private and public cloud infrastructures.
- Blowfish:
  - Best suited for **non-sensitive applications**, **edge computing**, and **IoT environments** where speed and low resource usage are critical.
  - Ideal for **small to medium-sized datasets** and applications that do not require the highest level of security.

CONCLUSION

- Twofish is generally the better choice for security-sensitive, large-scale hybrid cloud environments, offering stronger encryption and scalability for handling large datasets and compliance with regulatory standards.
- Blowfish is optimal for resource-constrained environments where speed and efficiency are prioritized over maximum security, making it ideal for smaller datasets and low-resource applications.
- The decision between the two depends on the specific needs of the hybrid cloud deployment, balancing security, performance, and scalability.

FUTURE ENHANCEMENT

- Exploring Hybrid Encryption Techniques:
  - Combining the strengths of Twofish and Blowfish with other encryption algorithms (e.g., AES) to enhance both security and performance in hybrid cloud environments.
- Quantum-Resistant Algorithms:
  - Researching quantum-resistant encryption algorithms to future-proof hybrid cloud systems against potential quantum computing threats.
- AI and ML Integration:
  - Leveraging Artificial Intelligence (AI) and Machine Learning (ML) to optimize encryption key management and detect security threats in real-time across hybrid cloud infrastructures.

REFERENCES

1. Aryotejo, G., & Kristiyanto, D. Y. (2018, May). Hybrid cloud: bridging of private and public cloud computing. In Journal of Physics: Conference Series (Vol. 1025, No. 1, p. 012091). IOP Publishing.

2. Tyagi, K., Yadav, S. K., & Singh, M. (2021, August). Cloud data security and various security algorithms. In Journal of Physics: Conference Series (Vol. 1998, No. 1, p. 012023). IOP Publishing.

3. Kaleem, M., Mushtaq, M. A., Jamil, U., Ramay, S. A., Khan, T. A., Patel, S., ... & Hussain, S. K. (2024). New Efficient Cryptographic Techniques For Cloud Computing Security. Migration Letters, 21(S11), 13-28.

4. Iavich, M., Gnatyuk, S., Jintcharadze, E., Polishchuk, Y., Fesenko, A., & Abisheva, A. (2019, July). Comparison and hybrid implementation of blowfish, twofish and rsa cryptosystems. In 2019 IEEE 2nd Ukraine Conference on Electrical and Computer Engineering (UKRCON) (pp. 970-974). IEEE.

5. Assa-Agyei, K., & Olajide, F. (2023). A Comparative study of Twofish, Blowfish, and advanced encryption standard for secured data transmission. International Journal of Advanced Computer Science and Applications, 14(3).

6. Santoso, K. I., Muin, M. A., & Mahmudi, M. A. (2020, April). Implementation of AES cryptography and twofish hybrid algorithms for cloud. In Journal of Physics: Conference Series (Vol. 1517, No. 1, p. 012099). IOP Publishing.

REFERENCES

7. Seth, B., Dalal, S., Le, D. N., Jaglan, V., Dahiya, N., Agrawal, A., ... & Verma, K. D. (2021). Secure Cloud Data Storage System Using Hybrid Paillier–Blowfish Algorithm. Computers, Materials & Continua, 67(1).

8. Thakur, A. J., & Jugele, R. N. SYMMETRIC ENCRYPTION: COUNTERING DATA PROTECTION AND DATA SECURITY IN CLOUD CRYPTOGRAPHY.

9. Kaur, J., & Garg, D. S. (2015). Security in cloud computing using hybrid of algorithms. International Journal of Engineering Research and General Science, 3(5), 300-305.

10. Hassan, O., & Al-Rawi, M. (2023). Comparative Study of Cloud Encryption Algorithms and Their Role in Securing Data Across Public, Private, and Hybrid Cloud Environments. Applied Research in Artificial Intelligence and Cloud Computing, 6(1), 110-120.

11. Balasubramanian, R., & Aramudhan, M. (2012). Security issues: public vs private vs hybrid cloud computing. International Journal of Computer Applications, 55(13).

12. Mohammad, O. K. J., Abbas, S., El-Horbaty, E. S. M., & Salem, A. B. M. (2013, December). A comparative study between modern encryption algorithms based on cloud computing environment. In 8th International Conference for Internet Technology and Secured Transactions (ICITST-2013) (pp. 531-535). IEEE.